

Supermarket Sales Analysis

Title: Supermarket Sales Analysis Using Python

Objectives:

1. Analyze supermarket sales data using Python.
2. Calculate the total sales.
3. Find the highest-selling product.
4. Calculate average sales.
5. Display products with sales greater than ₹1000.
6. Sort products by sales amount.

Tools used:

Python 3

Google Colab or Jupyter Notebook

Pandas Library

Requirements:

Install Pandas (if needed):

```
pip install pandas
```

Procedure:

1. Import the Pandas library.
2. Create a supermarket sales dataset.
3. Convert the data into a DataFrame.
4. Display the dataset.
5. Calculate total sales.
6. Calculate average sales.
7. Find the highest-selling product.
8. Display products with sales above ₹1000.
9. Sort the data by sales.
10. Save the results as a CSV file.

Program:

```
import pandas as pd
```

```
sales = {
```

```
    "Product": [
```

```
        "Rice",
```

```
        "Milk",
```

```
        "Sugar",
```

```
        "Oil",
```

```
        "Soap"
```

```
    ],
```

```
    "Sales": [
```

```
        1500,
```

```
        900,
```

```
        1200,
```

```
        1800,
```

```
        700
```

```
    ],
```

```
    "Quantity": [
```

```
        30,
```

```
        20,
```

```
        25,
```

```
        15,
```

```
        35
```

```
    ]
```

```
}
```

```
df = pd.DataFrame(sales)
```

```
print("Supermarket Sales Dataset")
```

```

print(df)

print("\nTotal Sales")

print(df["Sales"].sum())

print("\nAverage Sales")

print(df["Sales"].mean())

print("\nHighest Selling Product")

print(df.loc[df["Sales"].idxmax()])

print("\nProducts with Sales Above 1000")

print(df[df["Sales"] > 1000])

print("\nProducts Sorted by Sales")

sorted_sales = df.sort_values(by="Sales", ascending=False)

print(sorted_sales)

sorted_sales.to_csv("Supermarket_Sales_Analysis.csv", index=False)

print("\nProject Completed Successfully")

```

Expected Output:

Supermarket Sales Dataset

S.no	Product	Sales	Quantity
0	Rice	1500	30
1	Milk	900	20
2	Sugar	1200	25
3	Oil	1800	15
4	Soap	700	35

Total Sales

6100

Average Sales

1220.0

Highest Selling Product

Product	Oil
Sales	1800
Quantity	15

Products with Sales Above 1000

Product	Sales
Rice	1500
Sugar	1200
Oil	1800

Products Sorted by Sales

Product	Sales
Oil	1800
Rice	1500
Sugar	1200
Milk	900
Soap	700

Project Completed Successfully

Conclusion:

This project demonstrates how to analyze supermarket sales using Python and the Pandas library. It calculates the total and average sales, identifies the highest-selling product, filters products with sales above ₹1000, sorts the sales data, and saves the analyzed results to a CSV file.